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| <br>सत्यमेव जयते | भारत सरकार<br>दूरसंचार विभाग<br>दूरसंचार अभियांत्रिकी केंद्र<br><b>Government of India</b><br><b>Department of Telecommunications</b><br><b>Telecommunication Engineering Centre</b><br><b>K.L. Bhawan, Janpath, New Delhi- 110001</b> | <br>ISO<br>9001:2015 |
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**No. 26-1/2022-FA/TEC**

**Date: 31.07.2025**

**Subject: Minutes of 4<sup>th</sup> meeting of National Working Group (NWG)-15 corresponding to ITU-T Study Group (SG)-15 held on 29.07.2025.**

Fourth (4<sup>th</sup>) meeting of National Working Group (NWG)-15 corresponding to ITU-T Study Group 15 regarding “*Networks, Technologies, and Infrastructures for Transport, Access & home*”, reconstituted for the Study Period 2025-28, was held on 29.07.2025 under the chairmanship of Sh. Rakesh Desai, DDG(FA), TEC through video-conferencing. The list of participants is placed at Annexure A.

2. At the outset, Sh. Jasvir Singh Panesar, Director (FA) TEC & Convener of NWG-15, welcomed all the participants and briefly explained about the schedule of the second meeting of ITU-T SG15 being held at ITU-T HQ, Geneva from 13-24 October, 2025 and submission deadlines. He also briefed the participants about the agenda and further proceedings of the meeting.

3. Sh. Rakesh Desai, Chair/ NWG-15 & DDG (FA) TEC, welcomed all participants in general and particularly new contributors / academia viz. experts from SFO & IIT-Madras, TEC/DOT Officers- Mr Vineet and Ms. Kanika, Prof.Aneesh from IIT-Hyderabad. He briefly apprised the members about deliberations in the correspondence discussions and interim Rapporteur Group meetings (RGMs) during April to July 2025. Sh. Desai, in his opening remarks, highlighted that contributions presented by the Indian experts during the interim meetings in respect of G.fso (Q6/15) and G Supplement LLSOT(W)DM (Q2/15) were very well appreciated by the SG15 experts and now contributors have to follow-up with the Editor concerned for submitting additional information. He pointed out that last date of uploading the contribution on ITU-T portal is 30<sup>th</sup> September 2025 as per meeting notice dated 23<sup>rd</sup> July 2025 circulated by TSB for the forthcoming second meeting of SG15. He stated that as per SOP of DOT, NWG-15 has to finalise the contributions by 5<sup>th</sup> September 2025 and prior to this, the potential contributions need to be discussed in atleast next two meetings of NWG-15. He, therefore, requested the members to enhance their efforts for preparing contributions and further emphasized the focused efforts by the members for draft new work-items leading to ITU-T recommendations. He also requested the members to attend online correspondence discussion meetings called by the concerned Rapporteurs of SG15 from August 2025 onwards. The Chair also underlined the requirements of preparing contributions considering the scope of the respective question and considering the latest draft of the working documents / TDs ((Temporary documents)

in case of ongoing work-items.

After the introduction, Sh. Rakesh Desai explained the members on potential work-items for contributions that were circulated among the members and also requested members to actively participate in the discussions on various contributions.

**4. Discussions were held on various potential work-items and contributions presented by the NWG-15 members as under:-**

| <b>Sl. No.</b> | <b>Contribution/<br/>Potential Work Item<br/>Title and Question<br/>Number</b>                                               | <b>Contributors /<br/>Experts</b>                                                                                                                                                              | <b>Suggestions<br/>from<br/>Members /<br/>Way<br/>forward</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------|------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1              | Proposal for Draft New Recommendation on 'Fan-In Fan-Out devices (FIFO ) for MCF' (Q7/15)                                    | Experts from Academia from IIT-Madras viz. Prof. Deepa, Balaji Srinivasan, Mr. Sobhanbabu, Dr. Chandan Singh Yadav, Rana Pratap; Expert from SFO-Mr Samuel Varghese, Sh.Rakesh Desai from TEC. | <ul style="list-style-type: none"> <li>- Justification to be more elaborated</li> <li>- Parameter values to be defined.</li> <li>- Draft to be improved considering other suggestions of the members.</li> </ul>                                                                                                                                                                                                                                                                                                                     |
| 2              | Draft New Recommendation "G.smmcf " Characteristics a single-mode weakly-coupled multi-core optical fibre and cable" (Q5/15) | Prof. Deepa, Mr. Sobhanbabu and Dr. Chandan Singh Yadav from IIT-Madras, Sh.Arvind and Sh. Sudipta Bhaumik from STL, Sh.Rakesh Desai & Ms. Mudita Chandra from TEC, other experts of NWG-15.   | <ul style="list-style-type: none"> <li>- Inputs through contributions may be provided through ongoing correspondence discussions (Editor draft) w.r.to:- optical compatibility of G.652 and G.654 fiber; interoperability (e.g., core pitch) among the WC-MCFs with G.652 and G.654 fibres; appropriate geometrical specification; Collaboration and harmonization with IEC SC86A, etc.</li> <li>- India will submit test results on MCF.</li> <li>- Last date of the correspondence activity ends at 12 September, 2025.</li> </ul> |

| Sl. No. | Contribution/<br>Potential Work Item<br>Title and Question<br>Number                                                         | Contributors /<br>Experts                                                                                             | Suggestions<br>from<br>Members /<br>Way<br>forward                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| 3       | Addition of broad architectural diagram of the metro transport network in G.8310 (2020) Amd 2. (Q12/15)                      | Ms.Divya Sharma and Sh. Jainendra Bahadur from NSCS, Sh. Deepika Ambwani and Sh. Ashish Kumar Pandey from CDOT, Delhi | <ul style="list-style-type: none"> <li>- Latest amendment approved in March 2025 to be taken as baseline.</li> <li>- Revision proposed shall be incorporated in the G.8310 Amd.2 with proper justification in the main proposal.</li> <li>- Contributors may please attend Q11/15 interim meeting scheduled on 19<sup>th</sup> August 2025 on recommendation 'fgMTN' and explore to improve the contribution further.</li> <li>- Draft to be improved considering other suggestions of the members.</li> </ul>                                                                                                               |
| 4       | Discussion on the proposal for Architecture of Wavelength detector and tuning device to be used in Multi PON network (Q2/15) | Ms.Divya Sharma and Sh. Jainendra Bahadur from NSCS, Sh. Ashish Kumar Pandey from CDOT, Delhi                         | <ul style="list-style-type: none"> <li>- Contribution needs to be ascertained with the scope of Q2/15. Scope of Q6 and Q7/15 also needs to be looked into.</li> <li>- Ongoing work-item / contribution WD06-17 in Q6/15 regarding 'Self-tuning applications in G.698.4 (Multichannel bi-directional DWDM applications with port agnostic single-channel optical interfaces) and observations in Q6/15 report of interim meeting of June 2025 may please looked into regarding the tuning mechanisms developed at Q2 (G.989.x, etc).</li> <li>- Draft to be improved considering other suggestions of the members.</li> </ul> |

| Sl. No. | Contribution/<br>Potential Work Item<br>Title and Question<br>Number                                                                  | Contributors /<br>Experts                                                                                                                                                                                                                                        | Suggestions<br>from<br>Members /<br>Way<br>forward                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5       | Draft New Recommendation G.fso<br>“Terrestrial free space<br>optics for mobile<br>backhaul with short<br>reach interfaces”<br>(Q6/15) | Sh.Rakesh Desai &<br>Ms. Mudita Chandra<br>from TEC, Dr. A.<br>Paventhan and S<br>Kirubakaran from<br>ERNET, Sh. Gyan<br>Deep Verma, Mr<br>Abhishek Dixit /<br>Ms.Anjali Gupta /<br>Ms. Anupama Sharma<br>from IIT-Delhi, Mr<br>Hardik Soni from M/s<br>Navlife. | <ul style="list-style-type: none"> <li>- As agreed in Paris meeting (WD06-49), contributors have to furnish further information on remaining two main issues, one on fibre length in Clause 6.2.1 and the power outage probability calculation method description and second, Parameter values in Clause 8.</li> <li>- Draft to be improved considering other suggestions of the members.</li> <li>- Last date of the correspondence activity ends at 31 August 2025.</li> <li>- This work-item is targeted for consent at the next SG15 Plenary meeting in October 2025.</li> </ul> |
| 6       | L.nis-Practical<br>considerations for<br>network infrastructures<br>sharing (Q7/15)                                                   | Sh.Rakesh Desai &<br>Ms. Mudita Chandra<br>from TEC and Sh.<br>Rajesh Kumar Soni-<br>DDG (CS) DOT HQ,<br>Mr Abdul from<br>Airtel, Experts from<br>TSPs/IPs, other<br>experts of NWG-15.                                                                          | <ul style="list-style-type: none"> <li>- Inputs may be provided through ongoing correspondence discussions (Editor draft) w.r.to Passive components, Supporting infrastructures, Deployment consideration between shared and dedicated infrastructures, National experiences, etc</li> <li>- DOT vide OM dt.09.07.2025 on “Frame work for establishing Telecommunication Infrastructure in Building Development” prescribes that Telecommunication Infrastructure (TI) inside the</li> </ul>                                                                                         |

| Sl. No. | Contribution/<br>Potential Work Item<br>Title and Question<br>Number                                                                                                 | Contributors /<br>Experts                                                                                     | Suggestions<br>from<br>Members /<br>Way<br>forward                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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|         |                                                                                                                                                                      |                                                                                                               | <p>buildings / properties shall be shareable and interoperable. Hence, TSPs/ISPs/Ips may be requested to share their inputs in respect of the technical and practical requirements for infrastructure sharing.</p> <ul style="list-style-type: none"> <li>- Last date of the correspondence activity ends at 19 September, 2025.</li> </ul>                                                                                                                                                                                                                                                                                    |
| 7       | Draft New Recommendation G.dfos “Distributed fibre optic sensing system for terrestrial optical transmission system” (Q6/15)                                         | Experts from Industries (JMV LPS Ltd, STL), Sh. Vineet Malik, Director (Radio), TEC, other experts of NWG-15. | <ul style="list-style-type: none"> <li>- Contributor may please look into the latest working document on G.dfos-WD06-46 and also study other recommendations related to energy efficiency requirements in Optical systems considering the fact that optical transceivers are consuming very less energy.</li> <li>- Observations in Q6/15 report of interim meeting of June 2025 may please looked into on G.dfos.</li> <li>- Industry experts may consider to contribute on case studies for backscattering type sensing.</li> <li>- Last date of the correspondence activity ends at 31<sup>st</sup> August 2025.</li> </ul> |
| 8       | Draft Revision of Recommendation G.665 “Generic characteristics of Raman amplifiers and Raman amplified subsystems”- Adding new the new Appendix IV describing Raman | Experts from Academia and Industry (IIT-M, IIT-Hyderabad, M/s Tejas, etc), other members of NWG-15.           | <ul style="list-style-type: none"> <li>- Draft contribution may be prepared considering the ongoing work-item WD06-48.</li> <li>- This work-item is targeted for consent at the next SG15 Plenary meeting in October 2025.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                          |

| Sl. No. | Contribution/<br>Potential Work Item<br>Title and Question<br>Number                                                                                                                                              | Contributors /<br>Experts                                                                                                                                                                                           | Suggestions<br>from<br>Members /<br>Way<br>forward                                                                                                                                                                                                                                                                                                                                                                                                      |
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|         | amplifiers safety<br>mechanisms (Q6/15)                                                                                                                                                                           |                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 9       | G Supplement<br>LLSoT(W)DM: Low<br>Layer Split Mobile<br>Fronthaul over T(W)DM<br>PON- Revision G989.3<br>(NG-PON2) as a<br>Fronthaul Technology to<br>Support Coordinated<br>Multi-Point (CoMP) in<br>6G (Q2/15) | Prof. Goutam Das,<br>Preeti Samhita Pati,<br>Sourabh Bhandarkar<br>and Sourav Dutta<br>from IIT Kharagpur,<br>Atri Mukhopadhyay<br>from IIT<br>Bhubaneswar,<br>Sh.Rakesh Desai-<br>TEC, other members<br>of NWG-15. | <ul style="list-style-type: none"> <li>- Contributors may please pursue with the Editor of G Supplement LLSoT(W)DM for submitting additional information so that Indian proposals can be incorporated in the TD.</li> <li>- Q2/15 next RGM and a joint meeting of Q2,3 and 4/15 is scheduled on Sep 8-11 2025 and 26<sup>th</sup> August 2025 &amp; 16 September 2025 respectively. Deadline for contributions is Friday before the meeting.</li> </ul> |
| 10      | Draft Revised<br>supplement G.suppl.55<br>“Radio-over-fibre (RoF)<br>technologies and their<br>applications” (Q2/15)                                                                                              | Experts from IIT-M,<br>AOC testbed Industry<br>partners, CDOT, TEC<br>and other members of<br>NWG-15.                                                                                                               | <ul style="list-style-type: none"> <li>- Draft contribution may be prepared considering the working document ‘G.suppl.55 Rev 3-Editor Draft v2, 250618_D51’</li> <li>- Participation in the Q2/15 next RGM scheduled from September 8-11 2025. Deadline for contributions is Friday before the meeting.</li> </ul>                                                                                                                                      |
| 11      | Draft Revised<br>Recommendation L.361<br>“ID tag requirements for<br>infrastructure and<br>network elements<br>management” (Q7/15)                                                                                | NWG-15 Members                                                                                                                                                                                                      | <ul style="list-style-type: none"> <li>- Inputs may be provided through ongoing correspondence discussions on the Editor’s draft w.r.to the main revisions: - a) add a new clause 6.2.4 dealing with optical ID tag; b) add a new appendix V about an example of optical ID tag.</li> <li>- Last date of the correspondence activity ends at 20 September, 2025.</li> </ul>                                                                             |
| 12      | Draft New<br>Recommendation L.mct                                                                                                                                                                                 | NWG-15 Members                                                                                                                                                                                                      | <ul style="list-style-type: none"> <li>- Inputs may be provided through ongoing</li> </ul>                                                                                                                                                                                                                                                                                                                                                              |

| <b>Sl. No.</b> | <b>Contribution/<br/>Potential Work Item<br/>Title and Question<br/>Number</b>                                                                                             | <b>Contributors /<br/>Experts</b>                                            | <b>Suggestions<br/>from<br/>Members /<br/>Way<br/>forward</b>                                                                                                                                                                                                                    |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                | “Maintenance of conduits for telecommunication” (Q7/15)                                                                                                                    |                                                                              | correspondence discussions on the Editor’s draft w.r.to Underground conduit, Bridge-supported conduit, inspection schedules for conduits, etc.<br>- Last date of the correspondence activity ends at 18 <sup>th</sup> September 2025.                                            |
| 13             | Contributions during correspondence discussions of Q5/15 work-items viz. Revision of L.102, L.105, L.107, L.108, L.110, L.111, L.Suppl.39, New L.ocwp, L.Suppl.ne. (Q5/15) | NWG-15 Members                                                               | - Draft contribution may be prepared considering the respective working documents and considering the RGM report of Q5/15.                                                                                                                                                       |
| 14             | Potential work-item – proposed revised Recommendation L.403/L.37 “Optical branching components (non-wavelength selective)” (Q7/15)                                         | Experts from OEMs of Optical splitter, TEC officers, other members of NWG-15 | - Draft contribution may be prepared considering the latest developments in the Optical splitter technologies and working out gap analysis for L.403/L/37.                                                                                                                       |
| 15             | New ITU-T draft Recommendation G.AI_CPE- “AI-Enhanced Customer Premises Equipment for Intelligent In-Premises Networks” (Q3/15)                                            | Kanika Gambhir, DOT                                                          | - Justification for the proposed new work-item needs to be more elaborated and to be strengthened.<br>- Inputs through contribution may be submitted through RGM of Q3/15 or through correspondence discussions.<br>- Members need more explanations to understand the proposal. |

**The Chair thanked all the contributors for presenting their contributions before NWG-15 members. Contributors and NWG-15 members were further requested to**

**prepare afresh contribution(s) and / or improve the presented contribution(s) by 14<sup>th</sup> August 2025. Contributors and NWG-15 members were also requested to submit further inputs during correspondence discussions, adhering to the timelines prescribed by the Editor / Rapporteur of the respective work-item.**

5. Sh. Rakesh Desai once again requested the NWG-15 members, not having TIES account, to obtain TIES account by registering on ITU portal and also requested all members to start access of ITU-T SG15 documents / recommendations, study it for gap analysis and prepare contributions to address such gaps.

**6. The key action points emerged from the meeting are as under: -**

- i. Members / experts are requested to contribute on the work-items as discussed at para 4 above and any other work-items / new proposals related to the ITU-T SG15.
- ii. Members are requested to form a focused group to discuss and develop a contribution.
- iii. Members may study all the questions of ITU-T SG15 and come forward to increase number of Indian contributions in ITU-T SG15.
- iv. Members are requested to participate actively in the forthcoming correspondence discussion meetings of the SG15.

7. The meeting ended with the vote of thanks to the chair .



(Jasvir Singh Panesar)

Convener, NWG-15 & Director (FA), TEC

To,

All the participants / Members of the NWG-15.

Copy to: 1. Advisor (T), DOT HQ, New Delhi. for kind information .

2. Sr .DDG & Head, TEC for kind information .



**List of Participants**

1. Sh. Rakesh Desai, DDG (FA) & Chairman/DCC, TEC
2. Sh. Jasvir Singh Panesar, Director (FA) & Convener, TEC
3. Sh. Sudipta Bhaumik, Sterlite Technologies Ltd
4. Sh. M K Gupta, DDG SP DBN, Digital Bharat Nidhi (DBN) of DoT
5. Sh. Aneesh Sobhanan, IIT Hyderabad
6. Sh. Vineet Malik, Director (Radio), TEC
7. Sh. Ashish Kumar Pandey, CDOT Delhi
8. Ms. Shona Grover, TEC
9. Ms. Mudita Chandra, ADG (Tx), TEC
10. Ms. Kanika Gambhir, DoT
11. Sh. Chandan Singh Yadav, IIT Madras
12. Sh. Rana Pratap, IIT Madras, Academia
13. Sh. Jonaq Niveer Sharma, Indian Institute of Technology Madras
14. Sh. Sobhan Babu Chunduru, IITM
15. Dr. Partha Sarathi Das, Haldia institute of technology
16. Sh. Rishi Kapoor, EXFO
17. Sh. Srirama Murty Nistala, Aprecomm Pvt Ltd
18. Sh. Sudipta Bhaumik, Sterlite Technologies Ltd
19. Sh. Manjunath Harsha, SFO Technologies Pvt. Ltd.
20. Sh. Sudipto Das PHOENIX CONTACT, Phoenix Contact India Pvt Ltd
21. NSCS, New Delhi
22. Sh. Deo Pratap, AD (FA), TEC
23. Ms. Alka Kumari, Intern (FA), TEC

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